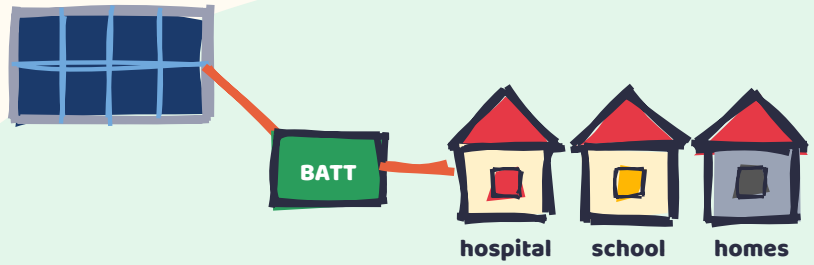
Tech Leaders · Class 23

Smart Microgrid Model

Sun in, power out, smart decisions in between.

Balance | Measure | Priority | Dashboard



Did you know?
Villages in Pakistan's mountains and Thar have been powered by small solar and hydro microgrids where the national grid doesn't reach.

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Let's think...

Why does load-shedding happen?

Lights out

At 8 pm everyone switches on fans, lights and TVs. If power stations can't produce that much, the grid must cut some areas off, or the whole grid could collapse. What if a smart controller decided **which** loads to cut, and kept the most important ones running?



Not enough power?

Bad wires?

A balance problem!

Today: you build a **solar microgrid model** with an ESP32 that measures power and switches loads by priority, with a live web dashboard.

Learning objectives

By the end of this class I can...

1	**How grids balance:** explain **generation, storage and load**, and why supply must equal demand every second	**Proof:** I can explain load-shedding as a balance problem
2	**Measure flows:** measure **solar input** and **load power** with two INA219 sensors on an ESP32	**Proof:** my Serial Monitor shows solar W, load W and battery V
3	**Priority control:** switch loads **on and off by priority** when energy is short	**Proof:** the 'hospital' LED stays on when the others shed
4	**Live dashboard:** serve a **web dashboard** from the ESP32 showing the grid state	**Proof:** my phone shows live watts and which loads are on

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Objective 1 · How grids balance · video

Video time!



What are Microgrids?

The B1M · about 4 minutes · youtu.be/lVlIPoV3Va4

Watch for...

1. What makes a microgrid different from the main grid?
2. Why can microgrids keep the lights on in a blackout?
3. Where are microgrids most useful?

Scan to watch

After the video: Which places in Pakistan would benefit most from a microgrid? Why?

Objective 1 · How grids balance

Supply must equal demand, every second

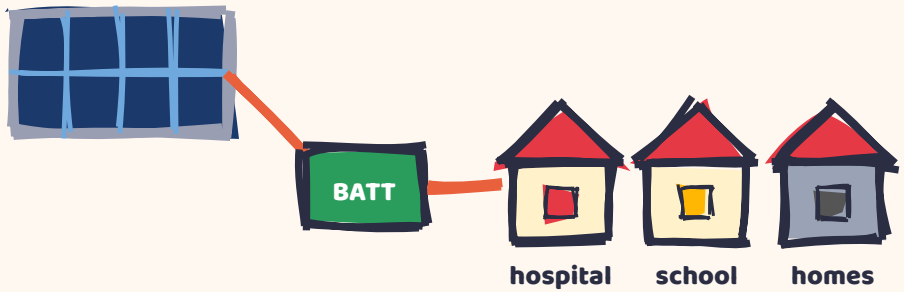
The idea

A grid has **generation** (solar, hydro, gas), **storage** (batteries, dams), and **loads** (homes, hospitals, factories).

Electricity is used the instant it's made. If demand > supply, voltage and frequency drop (Pakistan's grid runs at **50 Hz**); protection systems trip.

Fixes: add generation, use storage, or **shed load**. A **microgrid** is a small grid that can run on its own (islanded).

Example: a water tank with taps



Water flows in from a pipe (generation) and out through taps (loads). The tank is storage. If more taps run than the pipe supplies, the tank empties and someone must close a tap.

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Objective 2 · Measure flows

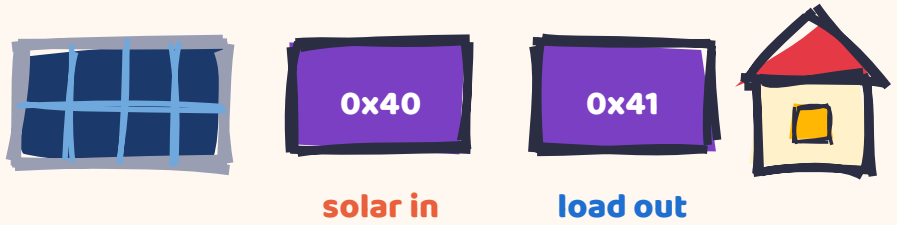
Measuring power flows

The idea

Two **INA219** sensors on one I²C bus with different **addresses** (0x40 and 0x41, set by solder jumper A0):
Solar sensor: power coming in.
Load sensor: power going out to the houses.

Net = solar − load. Positive: the battery charges. Negative: the battery drains.
Battery voltage tells its **state of charge** roughly (Li-ion: 4.2 V full, 3.3 V low).

Example: a bank statement



Money in (salary) and money out (spending). The balance tells you if you're saving or using savings. Energy accounting works the same way.

Objective 3 · Priority control

Priority-based load shedding

The idea

Give each load a **priority**: 1 = hospital (never cut if possible), 2 = school / water pump, 3 = homes' extra lights.

Rule: as battery voltage falls past thresholds, **cut lowest priority first**: V < 3.7 → cut priority 3. V < 3.5 → cut priority 2. V < 3.3 → cut everything (protect the battery).

Add **hysteresis**: reconnect only above a **higher** voltage (e.g. 3.8), or loads flicker on and off.

Example: a lifeboat



3.6 V

When supplies run low, a captain rations: medicine first, then water, then snacks. Fair rules decided in advance stop panic later.

Objective 4 · Live dashboard

A dashboard people can trust

The idea

The ESP32 runs a small **web server**. Phones on the same Wi-Fi open its IP and see: solar W, load W, battery V and %, and which loads are on.

Use **<meta http-equiv='refresh' content='2'>** for a simple auto-refresh, or JavaScript fetch() for smoother updates.

Show **why** a load was cut ('Low battery: homes paused'), so users trust the system instead of fighting it.

Example: a cricket scoreboard



A clear scoreboard (runs, overs, required rate) lets everyone understand the match instantly. A good dashboard does the same for energy.

Real world

Microgrids and smart grids



Village solar

Off-grid communities in Sindh and Balochistan



Micro-hydro

Mountain streams power villages in Gilgit-Baltistan



Hospitals

Solar + batteries keep critical equipment running in outages



Smart meters

Utilities shift and manage demand automatically



If your school ran on its own microgrid, which loads would be priority 1, 2 and 3? Why?

SDG 7 + 13

Affordable clean energy and climate action: microgrids help both, especially where the main grid is weak.

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Quick quiz

Grid logic

Question	Answer
1. Solar 3 W, load 5 W. Battery charging or draining?	
2. Why two different I²C addresses?	
3. Battery 3.6 V. Which loads are cut (using our rules)?	
4. What problem does hysteresis prevent?	
5. Pakistan's grid frequency?	

Answers

Cover! 1: draining 2 W · 2: so the ESP32 can talk to each sensor separately · 3: priority 3 only · 4: flickering on/off near a threshold · 5: 50 Hz

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Grid operator game
2. Our microgrid kit
3. Measure and manage by priority
4. The web dashboard
5. Run a day in the life
6. Grid log
7. Energy engineer challenges

Class game

Grid operator game

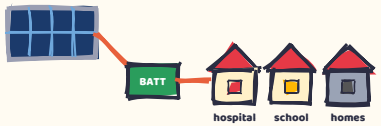
How to play

1. Teacher is 'the sun': calls supply each 'hour' (cards 0 to 10 units: night = 0!).
2. Teams are cities with loads: hospital 2, school 3, homes 4, factory 4 units. Battery holds 10.
3. Each hour: decide which loads run. Supply + battery must cover demand.
4. Run a full day (8 rounds). Points: hospital always on, fewest total cuts, battery never empty.

Then discuss

What rule did your team settle on?

Was it fair to everyone?



Big idea

Every grid needs rules for who gets power when there isn't enough.

Build it

Our microgrid kit



6 V solar panel
Generation



18650 + TP4056
Storage (charged by the panel)



ESP32 + 2 × INA219
Measure solar and load



3 logic MOSFETs
Switch the 3 loads (IRLZ44N or modules)



LED 'houses'
Card houses with LEDs



Phone
Views the dashboard

Lamp tip: indoors, a desk lamp close to the panel can stand in for the sun.

Battery safety
Lithium rules from YI-09 apply: no shorts, protected modules only, never charge unattended.

MAKE

Objective 3 · Priority control · Build step 1 of 3

Measure and manage by priority

```
#include <Wire.h>
#include <Adafruit_INA219.h>
Adafruit_INA219 solar(0x40), load(0x41);
const int LOAD_PIN[3] = {25, 26, 27};        // P1 hospital, P2 school, P3 homes
bool on[3] = {true, true, true};
float sW, lW, bV;

void manage() {
  // cut thresholds and (higher) reconnect thresholds = hysteresis
  const float cut[3] = {3.30, 3.50, 3.70}, back[3] = {3.50, 3.70, 3.90};
  for (int p = 0; p < 3; p++) {
    if (on[p] && bV < cut[p])   on[p] = false;
    if (!on[p] && bV > back[p]) on[p] = true;
    digitalWrite(LOAD_PIN[p], on[p]);
  }
}

void readGrid() {
  sW = solar.getPower_mW() / 1000.0;
  lW = load.getPower_mW() / 1000.0;
  bV = load.getBusVoltage_V();               // load sensor sits on the battery side
}
```

Two sensors

Bridge the A0 jumper on one INA219 to give it address 0x41.

Priority

Index 0 = hospital has the LOWEST cut voltage, so it's cut last.

Hysteresis

Reconnect only 0.2 V higher than the cut point: no flicker.

Objective 4 · Live dashboard · Build step 2 of 3

The web dashboard

```
#include <WiFi.h>
#include <WebServer.h>
WebServer server(80);
const char *names[3] = {"Hospital", "School", "Homes"};

void page() {
  String h = "<meta http-equiv='refresh' content='2'><h1>Microgrid</h1>";
  h += "<p>Solar: " + String(sW, 2) + " W<br>Load: " + String(lW, 2) + " W";
  h += "<br>Battery: " + String(bV, 2) + " V</p>";
  for (int p = 0; p < 3; p++)
    h += "<p>" + String(names[p]) + ": " + (on[p] ? "ON" : "PAUSED (low battery)") + "</p>";
  server.send(200, "text/html", h);
}

void setup() {
  Serial.begin(115200); Wire.begin(); solar.begin(); load.begin();
  for (int p = 0; p < 3; p++) pinMode(LOAD_PIN[p], OUTPUT);
  WiFi.begin("ClassHotspot", "password");
  while (WiFi.status() != WL_CONNECTED) delay(200);
  Serial.println(WiFi.localIP());
  server.on("/", page); server.begin();
}

void loop() {
  readGrid(); manage(); server.handleClient();
  static unsigned long t = 0;
  if (millis() - t > 1000) { t = millis(); Serial.printf("%.2f W in, %.2f W out, %.2f V\n", sW, lW, bV); }
}
```

Open it

Type the printed IP into a phone browser on the same Wi-Fi.

Why

The page says WHY a load is paused: trust matters.

Upgrade

Swap the refresh for fetch() + a JSON endpoint for smooth updates.

MAKE

Objective 3 · Build step 3 of 3

Run a day in the life

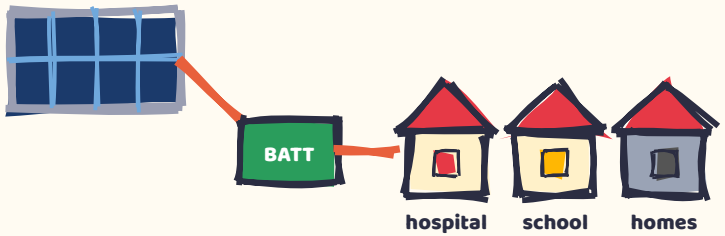
a Build the card village: 3 LED houses on MOSFETs, labelled P1 to P3.

b **Day**: lamp on the panel. Record solar W, load W, battery V for 5 minutes.

c **Night**: cover the panel. Watch the battery fall and loads shed in order.

d To speed up testing, temporarily raise the thresholds (e.g. +0.4 V) so shedding happens sooner. Put them back after!

e Screenshot the dashboard in each state for your report.



Check: Loads shed in priority order and reconnect without flicker.

Never connect the 6 V panel straight to the ESP32 or the battery: only through the TP4056.

Test like an engineer

Grid log

Test	Result	Notes
Day: solar / load / battery		
Night: time until P3 cut		min
P2 cut at		V
P1 ever cut?		
Flicker seen? (hysteresis OK?)		

Debug checklist

Both sensors read the same? Same address: check the A0 jumper and run an I²C scanner.
LEDs never switch? Logic-level MOSFET needed (IRLZ44N), and common GND.
Dashboard won't load? Phone and ESP32 must be on the same network.

Engineers test the edge cases: what happens at exactly the threshold, and when the sun comes back?

Level up!

Energy engineer challenges

Choose your challenge



Logging: log every minute to a CSV on SPIFFS and chart a full day in Python.



Forecast: use tomorrow's weather API (YI-30) to pre-charge before a cloudy day.



Fair rotation: rotate which homes are cut so no house is always last.



Demand response: dim non-critical LEDs with PWM instead of switching them off.

Think like an engineer

Is 'hospital first' always fair? Who decides priorities in a real community?

Boss level

Design the rules for a real microgrid for a village of 50 homes, a clinic and a school, with numbers.

Objective check

Can I really do it? Prove it.

01	Explain why supply must equal demand and how storage helps.	○ ○ ○
02	Show solar W, load W and battery V. Is the battery charging or draining?	○ ○ ○
03	Show loads shedding in priority order. Where is the hysteresis in your code?	○ ○ ○
04	Show the dashboard on a phone and explain one design choice.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Grid briefing: each team 'reports to the energy minister' with its data and one policy recommendation.

Show what you know

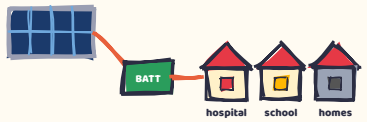
What did we learn?

- [x] Grids must balance supply and demand instantly
- [x] Storage covers gaps; shedding protects the system
- [x] Two INA219s measure power in and out
- [x] Priorities + hysteresis = fair, stable control
- [x] Dashboards should explain decisions

Exit ticket: the battery is at 3.6 V and falling. Which load goes off next, and when does it come back?

At home

During the next power cut at home, list what you'd keep on if you had a small battery.



Next class: Noise Pollution Analyser with FFT!